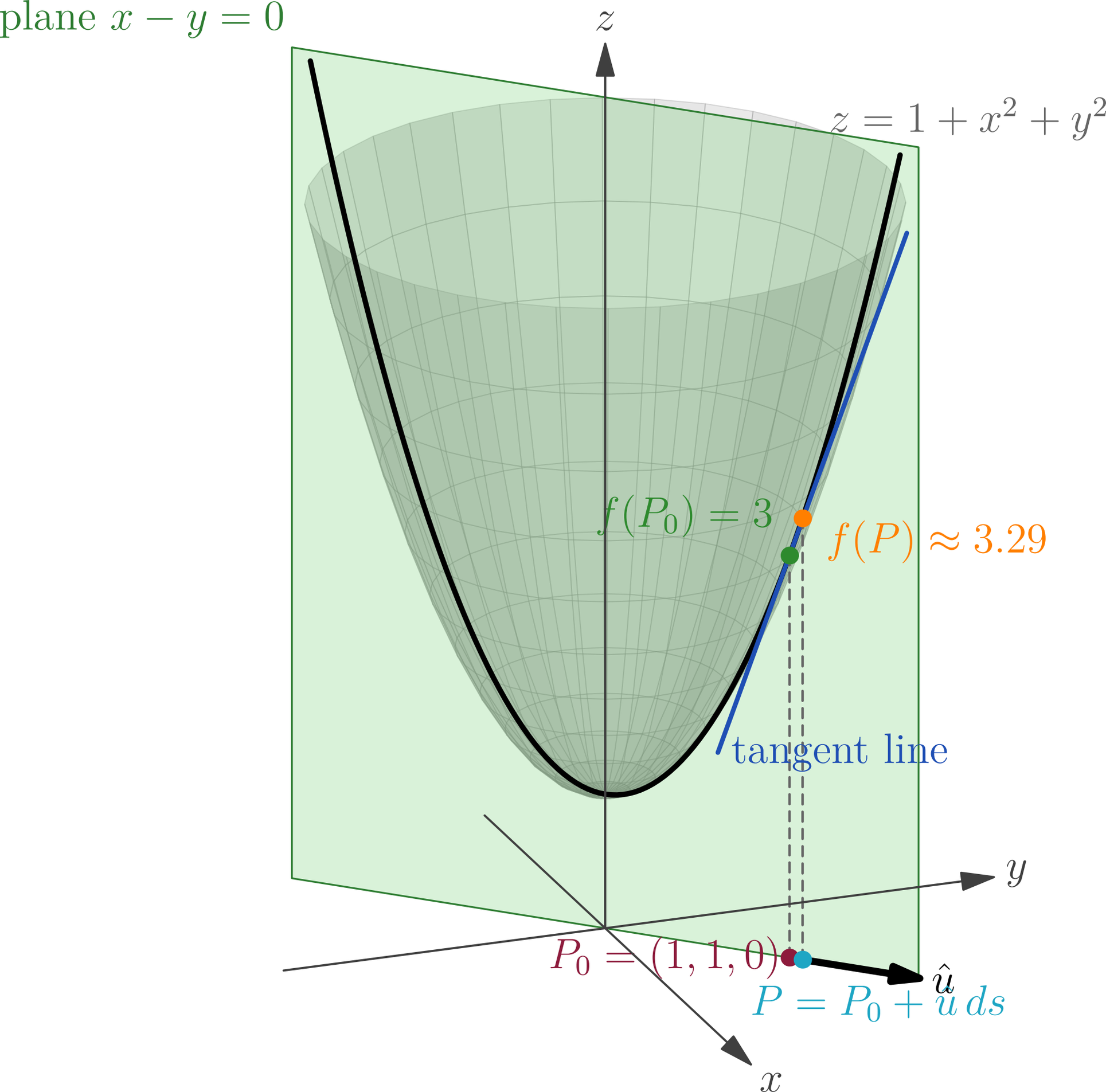
plane $x - y = 0$
z
$z = 1 + x^2 + y^2$
$f(P_0) = 3$
$f(P) \approx 3.29$
tangent line
y
$P_0 = (1, 1, 0)$
$\hat{u}$
$P = P_0 + \hat{u}\, ds$
x